Flatland

Edwin A. Abbott

Selected chapters: I-III and XIII-XV (1884). Original illustrations are retained; editorial endnotes are omitted.

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Part I: This World

I: Of the Nature of Flatland

I call our world Flatland, not because we call it so, but to make its nature clearer to you, my happy readers, who are privileged to live in Space.

Imagine a vast sheet of paper on which straight Lines, Triangles, Squares, Pentagons, Hexagons, and other Figures, instead of remaining fixed in their places, move freely about, on or in the surface, but without the power of rising above or sinking below it, very much like shadows—only hard and with luminous edges—and you will then have a pretty correct notion of my country and countrymen. Alas, a few years ago, I should have said “my universe”: but now my mind has been opened to higher views of things.

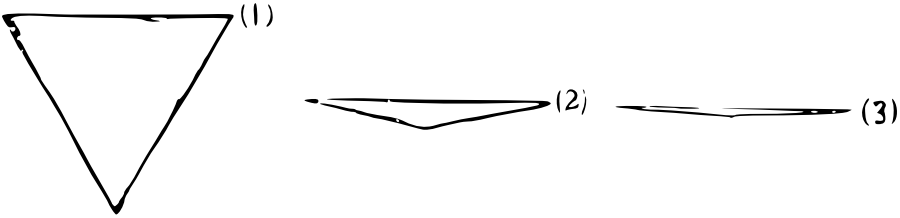
In such a country, you will perceive at once that it is impossible that there should be anything of what you call a “solid” kind; but I dare say you will suppose that we could at least distinguish by sight the Triangles, Squares, and other Figures, moving about as I have described them. On the contrary, we could see nothing of the kind, not at least so as to distinguish one Figure from another. Nothing was visible, nor could be visible, to us, except Straight Lines; and the necessity of this I will speedily demonstrate.

Place a penny on the middle of one of your tables in Space; and leaning over it, look down upon it. It will appear a circle.

But now, drawing back to the edge of the table, gradually lower your eye (thus bringing yourself more and more into the condition of the inhabitants of Flatland), and you will find the penny becoming more and more oval to your view, and at last when you have placed your eye exactly on the edge of the table (so that you are, as it were, actually a

Flatlander) the penny will then have ceased to appear oval at all, and will have become, so far as you can see, a straight line.

The same thing would happen if you were to treat in the same way a Triangle, or Square, or any other Figure cut out of pasteboard. As soon as you look at it with your eye on the edge of the table, you will find that it ceases to appear to you a Figure, and that it becomes in appearance a straight line. Take for example an equilateral Triangle—who represents with us a tradesman of the respectable class. Fig. 1 represents the tradesman as you would see him while you were bending over him from above; figs. 2 and 3 represent the tradesman, as you would see him if your eye were close to the level, or all but on the level of the table; and if your eye were quite on the level of the table (and that is how we see him in Flatland) you would see nothing but a straight line.



When I was in Spaceland I heard that your sailors have very similar experiences while they traverse your seas and discern some distant island or coast lying on the horizon. The far-off land may have bays, forelands, angles in and out to any number and extent; yet at a distance you see none of these (unless indeed your sun shines bright upon them revealing the projections and retirements by means of light and shade), nothing but a grey unbroken line upon the water.

Well, that is just what we see when one of our triangular or other acquaintances comes toward us in Flatland. As there is neither sun with us, nor any light of such a kind as to make shadows, we have none of the helps to the sight that you have in Spaceland. If our friend comes closer to us we see his line becomes larger; if he leaves us it becomes smaller:

but still he looks like a straight line; be he a Triangle, Square, Pentagon, Hexagon, Circle, what you will—a straight Line he looks and nothing else.

You may perhaps ask how under these disadvantageous circumstances we are able to distinguish our friends from one another: but the answer to this very natural question will be more fitly and easily given when I come to describe the inhabitants of Flatland. For the present let me defer this subject, and say a word or two about the climate and houses in our country.

II: Of the Climate and Houses in Flatland

As with you, so also with us, there are four points of the compass: North, South, East, and West.

There being no sun nor other heavenly bodies, it is impossible for us to determine the North in the usual way; but we have a method of our own. By a law of Nature with us, there is a constant attraction to the South; and, although in temperate climates this is very slight—so that even a woman in reasonable health can journey several furlongs northward without much difficulty—yet the hampering effect of the southward attraction is quite sufficient to serve as a compass in most parts of our earth. Moreover, the rain (which falls at stated intervals) coming always from the North, is an additional assistance; and in the towns we have the guidance of the houses, which of course have their sidewalls running for the most part North and South, so that the roofs may keep off the rain from the North. In the country, where there are no houses, the trunks of the trees serve as some sort of guide. Altogether, we have not so much difficulty as might be expected in determining our bearings.

Yet in our more temperate regions, in which the southward attraction is hardly felt, walking sometimes in a perfectly desolate plain where there have been no houses nor trees to guide me, I have been occasionally

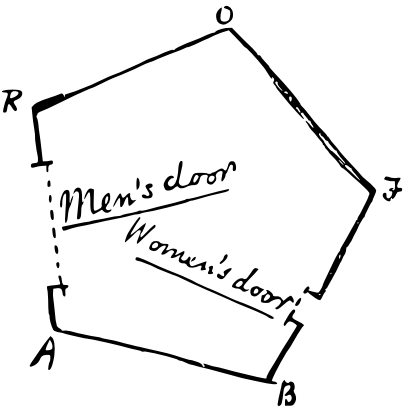
compelled to remain stationary for hours together, waiting till the rain came before continuing my journey. On the weak and aged, and especially on delicate females, the force of attraction tells much more heavily than on the robust of the male sex, so that it is a point of breeding, if you meet a lady in the street, always to give her the North side of the way—by no means an easy thing to do always at short notice when you are in rude health and in a climate where it is difficult to tell your North from your South.

Windows there are none in our houses: for the light comes to us alike in our homes and out of them, by day and by night, equally at all times and in all places, whence we know not. It was in old days, with our learned men, an interesting and oft-investigated question, “What is the origin of light?” and the solution of it has been repeatedly attempted, with no other result than to crowd our lunatic asylums with the would-be solvers. Hence, after fruitless attempts to suppress such investigations indirectly by making them liable to a heavy tax, the Legislature, in comparatively recent times, absolutely prohibited them. I—alas, I alone in Flatland—know now only too well the true solution of this mysterious problem; but my knowledge cannot be made intelligible to a single one of my countrymen; and I am mocked at—I, the sole possessor of the truths of Space and of the theory of the introduction of Light from the world of three Dimensions—as if I were the maddest of the mad! But a truce to these painful digressions: let me return to our houses.

The most common form for the construction of a house is five-sided or pentagonal, as in the annexed Figure. The two Northern sides RO , OF , constitute the roof, and for the most part have no doors; on the East is a small door for the women; on the West a much larger one for the men; the South side or floor is usually doorless.

Square and triangular houses are not allowed, and for this reason. The angles of a Square (and still more those of an equilateral Triangle), being much more pointed than those of a Pentagon, and the lines of inanimate

objects (such as houses) being dimmer than the lines of men and women, it follows that there is no little danger lest the points of a square or triangular house residence might do serious injury to an inconsiderate or perhaps absentminded traveller suddenly therefore, running against them: and as early as the eleventh century of our era, triangular houses were universally forbidden by law, the only exceptions being fortifications, powder-magazines, barracks, and other state buildings, which it is not desirable that the general public should approach without circumspection.



At this period, square houses were still everywhere permitted, though discouraged by a special tax. But, about three centuries afterwards, the law decided that in all towns containing a population above ten thousand, the angle of a Pentagon was the smallest house-angle that could be allowed consistently with the public safety. The good sense of the community has seconded the efforts of the Legislature; and now, even in the country, the pentagonal construction has superseded every other. It is only now and then in some very remote and backward agricultural district that an antiquarian may still discover a square house.

III: Concerning the Inhabitants of Flatland

The greatest length or breadth of a full grown inhabitant of Flatland may be estimated at about eleven of your inches. Twelve inches may be regarded as a maximum.

Our women are Straight Lines.

Our soldiers and lowest classes of workmen are Triangles with two equal sides, each about eleven inches long, and a base or third side so short (often not exceeding half an inch) that they form at their vertices a very sharp and formidable angle. Indeed when their bases are of the most degraded type (not more than the eighth part of an inch in size), they can hardly be distinguished from Straight Lines or women; so extremely pointed are their vertices. With us, as with you, these Triangles are distinguished from others by being called “Isosceles”; and by this name I shall refer to them in the following pages.

Our middle class consists of Equilateral or Equal-Sided Triangles.

Our professional men and gentlemen are Squares (to which class I myself belong) and Five-Sided Figures or Pentagons.

Next above these come the nobility, of whom there are several degrees, beginning at Six-Sided Figures, or Hexagons, and from thence rising in the number of their sides till they receive the honourable title of Polygonal, or many-sided. Finally when the number of the sides becomes so numerous, and the sides themselves so small, that the Figure cannot be distinguished from a Circle, he is included in the Circular or Priestly order; and this is the highest class of all.

It is a law of Nature with us that a male child shall have one more side than his father, so that each generation shall rise (as a rule) one step in the scale of development and nobility. Thus the son of a Square is a Pentagon; the son of a Pentagon, a Hexagon; and so on.

But this rule applies not always to the tradesmen, and still less often to the soldiers, and to the workmen; who indeed can hardly be said to deserve the name of human Figures, since they have not all their sides equal. With them therefore the law of Nature does not hold; and the son of an Isosceles (i.e. a Triangle with two sides equal) remains Isosceles still. Nevertheless, all hope is not shut out, even from the Isosceles, that his posterity may ultimately rise above his degraded condition. For, after a long series of military successes, or diligent and skilful labours, it is generally found that the more intelligent among the artisan and soldier classes manifest a slight increase of their third side or base, and a shrinkage of the two other sides. Intermarriages (arranged by the Priests) between the sons and daughters of these more intellectual members of the lower classes generally result in an offspring approximating still more to the type of the Equal-Sided Triangle.

Rarely—in proportion to the vast numbers of Isosceles births—is a genuine and certifiable Equal-Sided Triangle produced from Isosceles parents. Such a birth requires, as its antecedents, not only a series of carefully arranged intermarriages, but also a long, continued exercise of frugality and self-control on the part of the would-be ancestors of the coming Equilateral, and a patient, systematic, and continuous development of the Isosceles intellect through many generations.

The birth of a True Equilateral Triangle from Isosceles parents is the subject of rejoicing in our country for many furlongs around. After a strict examination conducted by the Sanitary and Social Board, the infant, if certified as Regular, is with solemn ceremonial admitted into the class of Equilaterals. He is then immediately taken from his proud yet sorrowing parents and adopted by some childless Equilateral, who is bound by oath never to permit the child henceforth to enter his former home or so much as to look upon his relations again, for fear lest the freshly developed organism may, by force of unconscious imitation, fall back again into his hereditary level.

The occasional emergence of an Equilateral from the ranks of his serf-born ancestors is welcomed, not only by the poor serfs themselves, as a gleam of light and hope shed upon the monotonous squalor of their existence, but also by the aristocracy at large; for all the higher classes are well aware that these rare phenomena, while they do little or nothing to vulgarize their own privileges, serve as a most useful barrier against revolution from below.

Had the acute-angled rabble been all, without exception, absolutely destitute of hope and of ambition, they might have found leaders in some of their many seditious outbreaks, so able as to render their superior numbers and strength too much even for the wisdom of the Circles. But a wise ordinance of Nature has decreed that, in proportion as the working-classes increase in intelligence, knowledge, and all virtue, in that same proportion their acute angle (which makes them physically terrible) shall increase also and approximate to the comparatively harmless angle of the Equilateral Triangle. Thus, in the most brutal and formidable of the soldier class—creatures almost on a level with women in their lack of intelligence—it is found that, as they wax in the mental ability necessary to employ their tremendous penetrating power to advantage, so do they wane in the power of penetration itself.

How admirable is this law of Compensation! And how perfect a proof of the natural fitness and, I may almost say, the divine origin of the aristocratic constitution of the States in Flatland! By a judicious use of this law of Nature, the Polygons and Circles are almost always able to stifle sedition in its very cradle, taking advantage of the irrepressible and boundless hopefulness of the human mind. Art also comes to the aid of law and order. It is generally found possible—by a little artificial compression or expansion on the part of the State physicians—to make some of the more intelligent leaders of a rebellion perfectly Regular, and to admit them at once into the privileged classes; a much larger number, who are still below the standard, allured by the prospect of being ultimately ennobled, are induced to enter the State Hospitals, where they

are kept in honourable confinement for life; one or two alone of the more obstinate, foolish, and hopelessly irregular are led to execution.

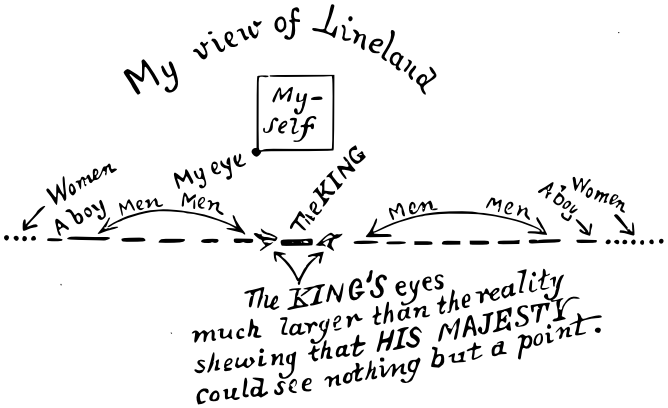
Then the wretched rabble of the Isosceles, planless and leaderless, are either transfixed without resistance by the small body of their brethren whom the Chief Circle keeps in pay for emergencies of this kind; or else more often, by means of jealousies and suspicions skilfully fomented among them by the Circular party, they are stirred to mutual warfare, and perish by one another’s angles. No less than one hundred and twenty rebellions are recorded in our annals, besides minor outbreaks numbered at two hundred and thirty-five; and they have all ended thus.

Part II: Other Worlds

XIII: How I Had a Vision of Lineland

It was the last day but one of the 1999th year of our era, and the first day of the Long Vacation. Having amused myself till a late hour with my favourite recreation of geometry, I had retired to rest with an unsolved problem in my mind. In the night I had a dream.

I saw before me a vast multitude of small Straight Lines (which I naturally assumed to be women) interspersed with other beings still smaller and of the nature of lustrous points—all moving to and fro in one and the same Straight Line, and, as nearly as I could judge, with the same velocity.



A noise of confused, multitudinous chirping or twittering issued from them at intervals as long as they were moving; but sometimes they ceased from motion, and then all was silence.

Approaching one of the largest of what I thought to be women, I accosted her, but received no answer. A second and a third appeal on my part were equally ineffectual. Losing patience at what appeared to me intolerable

rudeness, I brought my mouth into a position full in front of her mouth so as to intercept her motion, and loudly repeated my question, “Woman, what signifies this concourse, and this strange and confused chirping, and this monotonous motion to and fro in one and the same Straight Line?”

“I am no woman,” replied the small Line. “I am the Monarch of the world. But thou, whence intrudest thou into my realm of Lineland?” Receiving this abrupt reply, I begged pardon if I had in any way startled or molested his Royal Highness; and describing myself as a stranger I besought the King to give me some account of his dominions. But I had the greatest possible difficulty in obtaining any information on points that really interested me; for the Monarch could not refrain from constantly assuming that whatever was familiar to him must also be known to me and that I was simulating ignorance in jest. However, by persevering questions I elicited the following facts:

It seemed that this poor ignorant Monarch—as he called himself—was persuaded that the Straight Line which he called his Kingdom, and in which he passed his existence, constituted the whole of the world, and indeed the whole of Space. Not being able either to move or to see, save in his Straight Line, he had no conception of anything out of it. Though he had heard my voice when I first addressed him, the sounds had come to him in a manner so contrary to his experience that he had made no answer, “seeing no man,” as he expressed it, “and hearing a voice as it were from my own intestines.” Until the moment when I placed my mouth in his world, he had neither seen me, nor heard anything except confused sounds beating against—what I called his side, but what he called his inside or stomach ; nor had he even now the least conception of the region from which I had come. Outside his world, or Line, all was a blank to him; nay, not even a blank, for a blank implies Space; say, rather, all was nonexistent.

His subjects—of whom the small Lines were men and the Points women—were all alike confined in motion and eyesight to that single

Straight Line, which was their world. It need scarcely be added that the whole of their horizon was limited to a Point; nor could anyone ever see anything but a Point. Man, woman, child, thing—each was a Point to the eye of a Linelander. Only by the sound of the voice could sex or age be distinguished. Moreover, as each individual occupied the whole of the narrow path, so to speak, which constituted his universe, and no one could move to the right or left to make way for passers by, it followed that no Linelander could ever pass another. Once neighbours, always neighbours. Neighbourhood with them was like marriage with us. Neighbours remained neighbours till death did them part.

Such a life, with all vision limited to a Point, and all motion to a Straight Line, seemed to me inexpressibly dreary; and I was surprised to note the vivacity and cheerfulness of the King. Wondering whether it was possible, amid circumstances so unfavourable to domestic relations, to enjoy the pleasures of conjugal union, I hesitated for some time to question his Royal Highness on so delicate a subject; but at last I plunged into it by abruptly inquiring as to the health of his family. “My wives and children,” he replied, “are well and happy.”

Staggered at this answer—for in the immediate proximity of the Monarch (as I had noted in my dream before I entered Lineland) there were none but men—I ventured to reply, “Pardon me, but I cannot imagine how your Royal Highness can at any time either see or approach their Majesties, when there are at least half a dozen intervening individuals, whom you can neither see through, nor pass by? Is it possible that in Lineland proximity is not necessary for marriage and for the generation of children?”

“How can you ask so absurd a question?” replied the Monarch. “If it were indeed as you suggest, the Universe would soon be depopulated. No, no; neighbourhood is needless for the union of hearts; and the birth of children is too important a matter to have been allowed to depend upon such an accident as proximity. You cannot be ignorant of this. Yet

since you are pleased to affect ignorance, I will instruct you as if you were the veriest baby in Lineland. Know, then, that marriages are consummated by means of the faculty of sound and the sense of hearing.

“You are of course aware that every man has two mouths or voices—as well as two eyes—a bass at one and a tenor at the other of his extremities. I should not mention this, but that I have been unable to distinguish your tenor in the course of our conversation.” I replied that I had but one voice, and that I had not been aware that his Royal Highness had two. “That confirms my impression,” said the King, “that you are not a man, but a feminine monstrosity with a bass voice, and an utterly uneducated ear. But to continue.

“Nature having herself ordained that every man should wed two wives—”

“Why two?” asked I.

“You carry your affected simplicity too far,” he cried. “How can there be a completely harmonious union without the combination of the Four in One, viz. the bass and tenor of the man and the soprano and contralto of the two women?”

“But supposing,” said I, “that a man should prefer one wife or three?”

“It is impossible,” he said; “it is as inconceivable as that two and one should make five, or that the human eye should see a Straight Line.” I would have interrupted him; but he proceeded as follows:

“Once in the middle of each week a law of Nature compels us to move to and fro with a rhythmic motion of more than usual violence, which continues for the time you would take to count a hundred and one. In the midst of this choral dance, at the fifty-first pulsation, the inhabitants of the Universe pause in full career, and each individual sends forth his richest, fullest, sweetest strain. It is in this decisive moment that all our marriages are made. So exquisite is the adaptation of bass to treble, of

tenor to contralto, that oftentimes the loved ones, though twenty thousand leagues away, recognize at once the responsive note of their destined lover; and, penetrating the paltry obstacles of distance, Love unites the three. The marriage in that instant consummated results in a threefold male and female offspring which takes its place in Lineland."

"What! Always threefold?" said I. "Must one wife then always have twins?"

"Bass-voiced monstrosity! yes," replied the King. "How else could the balance of the sexes be maintained, if two girls were not born for every boy? Would you ignore the very alphabet of Nature?" He ceased, speechless for fury; and some time elapsed before I could induce him to resume his narrative.

"You will not, of course, suppose that every bachelor among us finds his mates at the first wooing in this universal Marriage Chorus. On the contrary, the process is by most of us many times repeated. Few are the hearts whose happy lot it is at once to recognize in each other's voices the partner intended for them by Providence, and to fly into a reciprocal and perfectly harmonious embrace. With most of us the courtship is of long duration. The wooer's voices may perhaps accord with one of the future wives, but not with both; or not, at first, with either; or the soprano and contralto may not quite harmonize. In such cases Nature has provided that every weekly Chorus shall bring the three lovers into closer harmony. Each trial of voice, each fresh discovery of discord, almost imperceptibly induces the less perfect to modify his or her vocal utterance so as to approximate to the more perfect. And after many trials and many approximations, the result is at last achieved. There comes a day at last, when, while the wonted Marriage Chorus goes forth from universal Lineland, the three far-off lovers suddenly find themselves in exact harmony, and, before they are awake, the wedded triplet is rapt vocally into a duplicate embrace; and Nature rejoices over one more marriage and over three more births."

XIV: How I Vainly Tried to Explain the Nature of Flatland

Thinking that it was time to bring down the Monarch from his raptures to the level of common sense, I determined to endeavour to open up to him some glimpses of the truth, that is to say of the nature of things in Flatland. So I began thus: “How does your Royal Highness distinguish the shapes and positions of his subjects? I for my part noticed by the sense of sight, before I entered your Kingdom, that some of your people are Lines and others Points, and that some of the Lines are larger—”

“You speak of an impossibility,” interrupted the King; “you must have seen a vision; for to detect the difference between a Line and a Point by the sense of sight is, as everyone knows, in the nature of things, impossible; but it can be detected by the sense of hearing, and by the same means my shape can be exactly ascertained. Behold me—I am a Line, the longest in Lineland, over six inches of Space—”

“Of Length,” I ventured to suggest.

“Fool,” said he, “Space is Length. Interrupt me again, and I have done.”

I apologized; but he continued scornfully, “Since you are impervious to argument, you shall hear with your ears how by means of my two voices I reveal my shape to my wives, who are at this moment six thousand miles seventy yards two feet eight inches away, the one to the North, the other to the South. Listen, I call to them.”

He chirruped, and then complacently continued: “My wives, at this moment receiving the sound of one of my voices, closely followed by the other, and perceiving that the latter reaches them after an interval in which sound can traverse 6.457 inches, infer that one of my mouths is 6.457 inches further from them than the other, and accordingly know my shape to be 6.457 inches. But you will of course understand that my wives do not make this calculation every time they hear my two voices.

They made it, once for all, before we were married. But they could make it at any time. And in the same way I can estimate the shape of any of my male subjects by the sense of sound."

"But how," said I, "if a man feigns a woman's voice with one of his two voices, or so disguises his Southern voice that it cannot be recognized as the echo of the Northern? May not such deceptions cause great inconvenience? And have you no means of checking frauds of this kind by commanding your neighbouring subjects to feel one another?" This of course was a very stupid question, for feeling could not have answered the purpose; but I asked with the view of irritating the Monarch, and I succeeded perfectly.

"What!" cried he in horror, "explain your meaning."

"Feel, touch, come into contact," I replied.

"If you mean by feeling ," said the King, "approaching so close as to leave no space between two individuals, know, stranger, that this offence is punishable in my dominions by death. And the reason is obvious. The frail form of a woman, being liable to be shattered by such an approximation, must be preserved by the State; but since women cannot be distinguished by the sense of sight from men, the law ordains universally that neither man nor woman shall be approached so closely as to destroy the interval between the approximator and the approximated.

"And indeed what possible purpose would be served by this illegal and unnatural excess of approximation which you call touching , when all the ends of so brutal and coarse a process are attained at once more easily and more exactly by the sense of hearing? As to your suggested danger of deception, it is nonexistent: for the voice, being the essence of one's being, cannot be thus changed at will. But come, suppose that I had the power of passing through solid things, so that I could penetrate my subjects, one after another, even to the number of a billion, verifying the size and distance of each by the sense of feeling : how much time and

energy would be wasted in this clumsy and inaccurate method! Whereas now, in one moment of audition, I take as it were the census and statistics, local, corporeal, mental and spiritual, of every living being in Lineland. Hark, only hark!"

So saying he paused and listened, as if in an ecstasy, to a sound which seemed to me no better than a tiny chirping from an innumerable multitude of lilliputian grasshoppers.

"Truly," replied I, "your sense of hearing serves you in good stead, and fills up many of your deficiencies. But permit me to point out that your life in Lineland must be deplorably dull. To see nothing but a Point! Not even to be able to contemplate a Straight Line! Nay, not even to know what a Straight Line is! To see, yet be cut off from those Linear prospects which are vouchsafed to us in Flatland! Better surely to have no sense of sight at all than to see so little! I grant you I have not your discriminative faculty of hearing; for the concert of all Lineland which gives you such intense pleasure, is to me no better than a multitudinous twittering or chirping. But at least I can discern, by sight, a Line from a Point. And let me prove it. Just before I came into your kingdom, I saw you dancing from left to right, and then from right to left, with seven men and a woman in your immediate proximity on the left, and eight men and two women on your right. Is not this correct?"

"It is correct," said the King, "so far as the numbers and sexes are concerned, though I know not what you mean by 'right' and 'left,' But I deny that you saw these things. For how could you see the Line, that is to say the inside, of any man? But you must have heard these things, and then dreamed that you saw them. And let me ask what you mean by those words 'left' and 'right,' I suppose it is your way of saying Northward and Southward."

"Not so," replied I; "besides your motion of Northward and Southward, there is another motion which I call from right to left."

King. Exhibit to me, if you please, this motion from left to right.

I. Nay, that I cannot do, unless you could step out of your Line altogether.

King. Out of my Line? Do you mean out of the world? Out of Space?

I. Well, yes. Out of your world. Out of your Space. For your Space is not the true Space. True Space is a Plane; but your Space is only a Line.

King. If you cannot indicate this motion from left to right by yourself moving in it, then I beg you to describe it to me in words.

I. If you cannot tell your right side from your left, I fear that no words of mine can make my meaning clear to you. But surely you cannot be ignorant of so simple a distinction.

King. I do not in the least understand you.

I. Alas! How shall I make it clear? When you move straight on, does it not sometimes occur to you that you could move in some other way, turning your eye round so as to look in the direction towards which your side is now fronting? In other words, instead of always moving in the direction of one of your extremities, do you never feel a desire to move in the direction, so to speak, of your side?

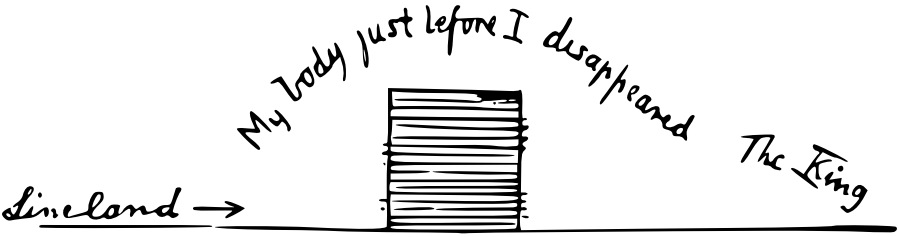
King. Never. And what do you mean? How can a man's inside "front" in any direction? Or how can a man move in the direction of his inside?

I. Well then, since words cannot explain the matter, I will try deeds, and will move gradually out of Lineland in the direction which I desire to indicate to you.

At the word I began to move my body out of Lineland. As long as any part of me remained in his dominion and in his view, the King kept exclaiming, "I see you, I see you still; you are not moving." But when I

had at last moved myself out of his Line, he cried in his shrillest voice, “She is vanished; she is dead.”

“I am not dead,” replied I; “I am simply out of Lineland, that is to say, out of the Straight Line which you call Space, and in the true Space, where I can see things as they are. And at this moment I can see your Line, or side—or inside as you are pleased to call it; and I can see also the men and women on the North and South of you, whom I will now enumerate, describing their order, their size, and the interval between each.”



When I had done this at great length, I cried triumphantly, “Does that at last convince you?” And, with that, I once more entered Lineland, taking up the same position as before.

But the Monarch replied, “If you were a man of sense—though, as you appear to have only one voice I have little doubt you are not a man but a woman—but, if you had a particle of sense, you would listen to reason. You ask me to believe that there is another Line besides that which my senses indicate, and another motion besides that of which I am daily conscious. I, in return, ask you to describe in words or indicate by motion that other Line of which you speak. Instead of moving, you merely exercise some magic art of vanishing and returning to sight; and instead of any lucid description of your new world, you simply tell me the numbers and sizes of some forty of my retinue, facts known to any child in my capital. Can anything be more irrational or audacious? Acknowledge your folly or depart from my dominions.”

Furious at his perversity, and especially indignant that he professed to be ignorant of my sex, I retorted in no measured terms, “Besotted being! You think yourself the perfection of existence, while you are in reality the most imperfect and imbecile. You profess to see, whereas you can see nothing but a Point! You plume yourself on inferring the existence of a Straight Line; but I can see Straight Lines, and infer the existence of Angles, Triangles, Squares, Pentagons, Hexagons, and even Circles. Why waste more words? Suffice it that I am the completion of your incomplete self. You are a Line, but I am a Line of Lines, called in my country a Square: and even I, infinitely superior though I am to you, am of little account among the great nobles of Flatland, whence I have come to visit you, in the hope of enlightening your ignorance.”

Hearing these words the King advanced towards me with a menacing cry as if to pierce me through the diagonal; and in that same moment there arose from myriads of his subjects a multitudinous war-cry, increasing in vehemence till at last methought it rivalled the roar of an army of a hundred thousand Isosceles, and the artillery of a thousand Pentagons. Spellbound and motionless, I could neither speak nor move to avert the impending destruction; and still the noise grew louder, and the King came closer, when I awoke to find the breakfast-bell recalling me to the realities of Flatland.

XV: Concerning a Stranger from Spaceland

From dreams I proceed to facts.

It was the last day of the 1999th year of our era. The pattering of the rain had long ago announced nightfall; and I was sitting in the company of my wife, musing on the events of the past and the prospects of the coming year, the coming century, the coming Millennium.

My four sons and two orphan Grandchildren had retired to their several apartments; and my wife alone remained with me to see the old Millennium out and the new one in.

I was rapt in thought, pondering in my mind some words that had casually issued from the mouth of my youngest grandson, a most promising young Hexagon of unusual brilliancy and perfect angularity. His uncles and I had been giving him his usual practical lesson in Sight Recognition, turning ourselves upon our centres, now rapidly, now more slowly, and questioning him as to our positions; and his answers had been so satisfactory that I had been induced to reward him by giving him a few hints on arithmetic, as applied to geometry.

Taking nine Squares, each an inch every way, I had put them together so as to make one large Square, with a side of three inches, and I had hence proved to my little grandson that—though it was impossible for us to see the inside of the Square—yet we might ascertain the number of square inches in a Square by simply squaring the number of inches in the side: "and thus," said I, "we know that 3 2 , or 9, represents the number of square inches in a Square whose side is 3 inches long."

The little Hexagon meditated on this a while and then said to me; "But you have been teaching me to raise numbers to the third power: I suppose 3 3 must mean something in geometry; what does it mean?"

"Nothing at all," replied I, "not at least in geometry; for geometry has only Two Dimensions." And then I began to show the boy how a Point by moving through a length of three inches makes a Line of three inches, which may be represented by 3; and how a Line of three inches, moving parallel to itself through a length of three inches, makes a Square of three inches every way, which may be represented by 3 2 .

Upon this, my grandson, again returning to his former suggestion, took me up rather suddenly and exclaimed, "Well, then, if a Point by moving three inches, makes a Line of three inches represented by 3; and if a

straight Line of three inches, moving parallel to itself, makes a Square of three inches every way, represented by 3 2 ; it must be that a Square of three inches every way, moving somehow parallel to itself (but I don't see how) must make something else (but I don't see what) of three inches every way—and this must be represented by 3 3 ."

"Go to bed," said I, a little ruffled by this interruption: "if you would talk less nonsense, you would remember more sense."

So my grandson had disappeared in disgrace; and there I sat by my wife's side, endeavouring to form a retrospect of the year 1999 and of the possibilities of the year 2000, but not quite able to shake off the thoughts suggested by the prattle of my bright little Hexagon. Only a few sands now remained in the half-hour glass. Rousing myself from my reverie I turned the glass Northward for the last time in the old Millennium; and in the act, I exclaimed aloud, "The boy is a fool."

Straightway I became conscious of a presence in the room, and a chilling breath thrilled through my very being. "He is no such thing," cried my wife, "and you are breaking the Commandments in thus dishonouring your own grandson." But I took no notice of her. Looking round in every direction I could see nothing; yet still I felt a presence, and shivered as the cold whisper came again. I started up. "What is the matter?" said my wife, "there is no draught; what are you looking for? There is nothing." There was nothing; and I resumed my seat, again exclaiming, "The boy is a fool, I say; 3 3 can have no meaning in geometry." At once there came a distinctly audible reply, "The boy is not a fool; and 3 3 has an obvious Geometrical meaning."

My wife as well as myself heard the words, although she did not understand their meaning, and both of us sprang forward in the direction of the sound. What was our horror when we saw before us a Figure! At the first glance it appeared to be a woman, seen sideways; but a moment's observation showed me that the extremities passed into dimness too rapidly to represent one of the female sex; and I should have

thought it a Circle, only that it seemed to change its size in a manner impossible for a Circle or for any regular Figure of which I had had experience.

But my wife had not my experience, nor the coolness necessary to note these characteristics. With the usual hastiness and unreasoning jealousy of her sex, she flew at once to the conclusion that a woman had entered the house through some small aperture. “How comes this person here?” she exclaimed, “you promised me, my dear, that there should be no ventilators in our new house.”

“Nor are there any,” said I; “but what makes you think that the Stranger is a woman? I see by my power of Sight Recognition—”

“Oh, I have no patience with your Sight Recognition,” replied she, “ ‘Feeling is believing’ and ‘A Straight Line to the touch is worth a Circle to the sight’ ”—two proverbs, very common with the frailer sex in Flatland.

“Well,” said I, for I was afraid of irritating her, “if it must be so, demand an introduction.” Assuming her most gracious manner, my wife advanced towards the Stranger, “Permit me, Madam, to feel and be felt by—” then, suddenly recoiling, “Oh! it is not a woman, and there are no angles either, not a trace of one. Can it be that I have so misbehaved to a perfect Circle?”

“I am indeed, in a certain sense a Circle,” replied the voice, “and a more perfect Circle than any in Flatland; but to speak more accurately, I am many Circles in one.” Then he added more mildly, “I have a message, dear Madam, to your husband, which I must not deliver in your presence; and, if you would suffer us to retire for a few minutes—” But my wife would not listen to the proposal that our august visitor should so incommode himself, and assuring the Circle that the hour of her own retirement had long passed, with many reiterated apologies for her recent indiscretion, she at last retreated to her apartment.

I glanced at the half-hour glass. The last sands had fallen. The third Millennium had begun.